

Quiz 7A

Filename: No File Submission

Instructions

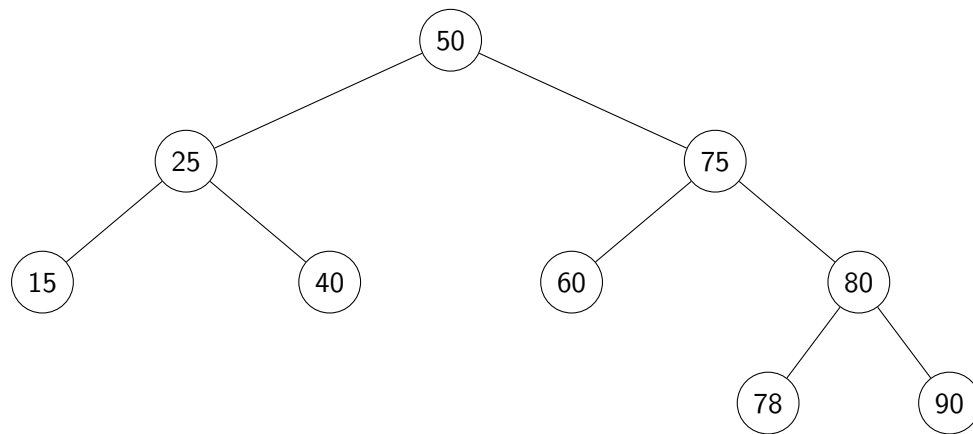
All answers for Quiz 7A must be submitted on Webcourses. On the other hand, answers for Quiz 7B must be submitted on Gradescope. Before proceeding with the quiz on Webcourses, you are expected to first work through and answer the questions here and on paper. Only once you have completed your work should you proceed to Webcourses and submit your responses.

Please note that some questions may not require you to submit your final written answers; however, you are still expected to fully solve every question on your own. As you work through the problems, you must time yourself. Each question is designed to be completed within a reasonable amount of time. Some questions may take approximately 1 minute, others around 3 minutes, but no question should take more than 5 minutes.

Important: Do not use a calculator when answering the following questions.

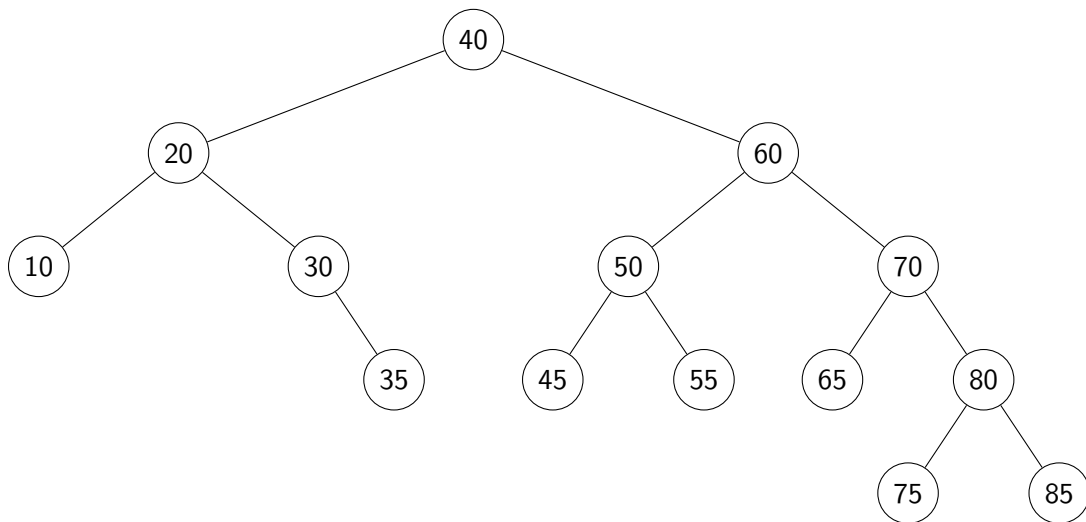
TODO 1 (15 points)

Insert the value 85 into the AVL tree shown below. If the insertion causes the tree to become unbalanced, perform the necessary rotations and redraw it after rebalancing.



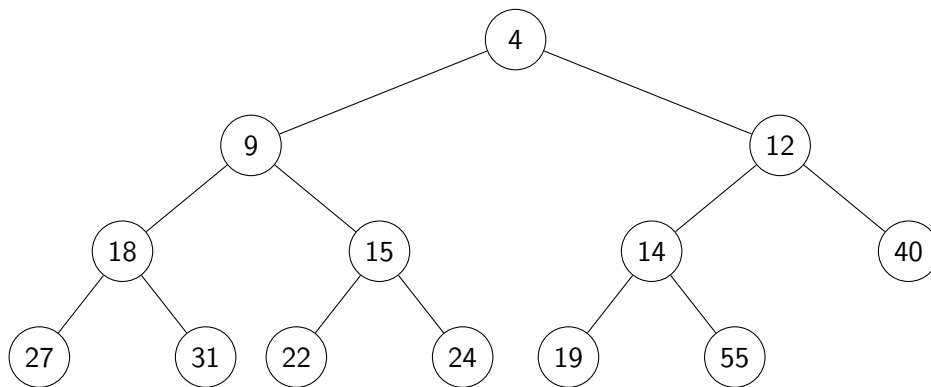
TODO 2 (20 points)

Delete the value 10 from the AVL tree shown below. If the deletion makes the tree unbalanced, perform the necessary rotations and redraw it after rebalancing.



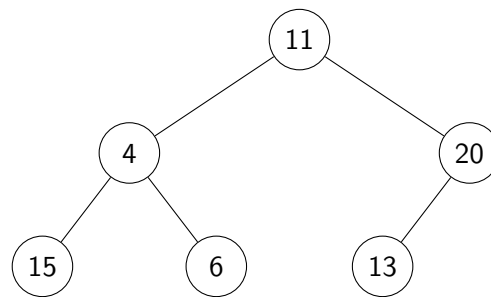
TODO 3 (10 points)

Consider the following min-heap. Remove the highest-priority element (i.e., the minimum value), then redraw the heap after fixing it.



TODO 4 (10 points)

Consider the following complete binary tree. Convert it into a max-heap using the build-heap procedure, then draw the resulting max-heap.



TODO 5 (15 points)

Part 1

Draw the trie obtained by inserting the following words, in the given order, into an initially empty trie. After constructing the trie, determine the total number of nodes, including the root.

polter, pol, pond, to, top, torn, ton

Part 2

Delete the word `polter` from the trie. Determine the total number of nodes remaining, including the root.

TODO 6 (20 points)

Consider a hash table of size $m = 13$ that uses **quadratic probing** for collision resolution. The hash function is defined as:

$$f(x) = (3x + 2) \% 13$$

Hint: The symbol % denotes the modulo operation, which returns the remainder after division.

Insert the following values into the hash table in the given order:

4, 1, 6, 3, 13, 17, 7

Index	0	1	2	3	4	5	6	7	8	9	10	11	12
Value													

TODO 7 (10 points)

Convert each of the following values from one base to another.

$$(9C)_{16} = \underline{\hspace{2cm}}_8$$

$$(117)_{10} = \underline{\hspace{2cm}}_2$$